



University of Perpetual Help System Laguna

City of Biñan, Laguna, Philippines, 4024
(+63)2779-5310 • 544-5150 • 544-5161
www.uphsl.edu.ph



College of Computer Studies

Final Activity

Business Intelligence Data Visualization LEC Data Visualization LAB

Group Capstone Project (Integrated DRRM Business Intelligence Dashboard)

Submitted by:

Funelas, Allan John
Rivera, France Raphael
Torculas, Richard

Submitted to:

Dr. Eliza Mapanoo

May 5, 2026



1. Introduction and BI context

Our project is a Business Intelligence dashboard for the City of Biñan, Laguna. Biñan is a city that is very prone to flooding because it is located right next to Laguna de Bay. Because of this, the city needs a better way to monitor flood incidents and rescue resources.

The target users for this dashboard are the DRRMO Officers and City Planners of Biñan. We classified this as an Operational and Tactical BI Dashboard. It is operational because it helps staff monitor the daily activities and incidents as they happen. It is tactical because it helps the leaders decide where to put their rescue teams and check if the evacuation centers are already full. The time horizon is mostly daily and periodic, focusing on current emergencies and short-term planning.

2. Data used in the system

We used four main datasets that we created based on the real barangays and population of Biñan City.

Dataset	Description	Key Fields	Used For
barangays.csv	Barangay-level hazard and population data	Barangay, HazardType, HazardLevel, Population, Latitude, Longitude	Risk scoring and hazard mapping
facilities.csv	Facilities with type and capacity	FacilityID, Name, Barangay, Type, Capacity, Latitude, Longitude	Capacity analysis and shelters
rescue_teams.csv	Team deployment and specialty data	TeamID, Name, BaseBarangay, Specialty, OnDuty/Availability, Latitude, Longitude	Readiness and deployment check
incidents.csv	Incident details and affected counts	IncidentID, DateTime, Barangay, Type, Severity,	Trend analysis and impact logs



3. Data dictionary, dimensions, and measures

We categorized our data into dimensions and measures to make the analysis easier:

Field	Data Type	Classification	Meaning	Used In
Month	Text	Dimension	Month of the event	Trend chart filter
Barangay	Text	Dimension	Location in Biñan	Mapping and ranking
Severity	Text	Dimension	Danger level	Severity pie chart
NumAffected	Integer	Measure	Number of people	Affected population KPI
Capacity	Integer	Measure	Max shelter space	Gap analysis chart
RiskScore	Float	Measure	Computed risk level	Prioritizing barangays

4. Data processing and ETL

For our data processing, we followed the BI concept of ETL (Extract, Transform, Load). In the Extraction phase, we used the Pandas library to load all four CSV files into our Jupyter Notebook.

During the cleaning and transformation phase, we found a few data issues. First, we noticed that some records in the 'NumAffected' column were blank or missing. We handled this by replacing the missing values with the median number of affected people for that specific type of incident. We also fixed some consistency issues with the DateTime column by converting it from a simple text string into a proper DateTime object. This was a very important step because it allowed us to extract the 'Month' and 'Year' for our trend analysis.

We also derived a new metric called the 'Risk Score' to help the DRRMO prioritize their response. We created this by assigning a numerical weight to the Hazard Level (Critical = 4, High = 3, Medium = 2, Low = 1) and then multiplying it by the population of the barangay. This transformed our raw data into a strategic insight that ranks the most vulnerable areas in Biñan.



5. KPIs and metrics

Our dashboard has 10 main KPIs to give a quick summary of the city's status:

KPI / Metric	Source Field	Update Frequency	Supports What Decision?
Total Incidents	Incidents count	Real-time	Monitors current workload
Total Affected	Sum(NumAffected)	Real-time	Checks the scale of impact
Available Teams	Status column	Real-time	Identifies who can be deployed
Total Capacity	Sum(Capacity)	Periodic	Checks total evacuation space
Highest Risk Brgy	RiskScore	Periodic	Helps in long-term planning
Evacuation Sites	Count(Facilities)	Periodic	Shows how many shelters are open
Water Rescue Teams	Specialty	Real-time	Important for flood response
Critical Incidents	Severity	Real-time	Highlights urgent emergencies
Occupancy Rate	Occupants/Capacity	Real-time	Prevents overcrowding in sites
Need Resupply	ResourcesAvailable	Real-time	Shows sites low on food/water

6. Dashboard design and explanation of visuals

We designed our charts to answer specific questions for the city responders:

- 1. Incident Trend Over Time:** This line chart answers 'Are incidents increasing?'. It uses the Month and Incident count. It helps the city prepare for seasonal floods.
- 2. Incident Types:** This bar chart shows which problems happen most often, like floods or fires.
- 3. Risk Ranking:** This bar chart uses the Risk Score to rank barangays. It tells the city where to focus its flood mitigation budget.
- 4. Gap Analysis:** This chart compares affected people vs. available capacity. It helps the city decide if they need to open more evacuation sites.



5. Severity Pie Chart: This shows the ratio of Critical vs. Low incidents. It gives an immediate idea of the overall threat level.

6. Interactive Map: We used Folium to plot every incident and facility. We used custom icons for hospitals and schools so the user can easily find help.

A key operational feature is the interactive data entry forms. The staff can log new incidents and add new facilities directly in the notebook. When they submit the form, the records are saved, and the dashboard updates automatically.

7. Interaction and analytical navigation

Our dashboard supports multidimensional analysis through interactive filters. The users can perform the following OLAP operations:

Slice: Filtering the dashboard to show only the 'Flood' incidents.

Dice: Filtering to show 'Critical' Floods during a specific 'Month'.

Drill-down: Moving from the total KPI counts down to the exact street address on the interactive map or the details table.

Roll-up: Aggregating individual incident reports into a monthly trend view to see the big picture.

8. DRRM insights and recommendations

From our analysis, we found that coastal barangays like Malaban and De La Paz are consistently at high risk. The gap analysis also shows that these areas often have overcrowded evacuation centers. Recommendation: The city should invest in bigger evacuation centers in these coastal areas. We also recommend placing specialized water rescue teams permanently in these barangays during the months of July to October.

10. Limitations and Improvements

The current system uses simulated data that we made as realistic as possible for Biñan City. In the future, we can improve this by connecting it to a live weather database or an automated SMS alert system. We can also add more dimensions like weather, humidity, and rainfall data to make the predictions better.